

Software Requirements Specification

ExpenseMate – Personal Expense Manager

Software Construction, Assignment 1 | Phase 1: Problem Definition & Requirements | Version 1.0, 27 September 2026

1. Introduction

Purpose. Defines what ExpenseMate must do and how well. It is the reference for design, construction and testing.

Problem. People track money in notebooks or scattered spreadsheets, so they cannot see where it goes and notice overspending too late.

Scope. Accounts, categories, income and expense records, monthly budgets with alerts, analytics and CSV import/export. Out of scope: bank sync, shared accounts and mobile apps. Multi-currency and cloud backup are planned for Phase 5.

Terms. *Transaction*: one income or expense. *Category*: a label such as Food. *Budget*: a monthly limit for one category. *Budget alert*: a warning at 80% of a budget and when it is exceeded.

2. Overall Description

Product. A client–server web app that runs locally: React client, Node.js/Express REST API, SQLite database via Prisma.

Users. One user type: a person managing their own money. Each account is private.

Environment. Windows, macOS or Linux with Node.js 20+; a current Chrome, Edge or Firefox; screens from 360 px wide.

Constraints. Amounts are in Pakistani Rupees and stored as whole paisa. CSV files use the columns *date*, *type*, *category*, *amount*, *note* (dates as YYYY-MM-DD).

3. Functional Requirements

ID	Requirement	Priority
FR-01	Sign up with name, email and a password of at least 8 characters; reject an email already in use.	High
FR-02	Create starter categories for every new account.	Medium
FR-03	Log in and log out; a session lasts up to 30 days.	High
FR-04	Show and change only the logged-in user's data.	High
FR-05	Add and delete categories; names are unique per user; a category in use cannot be deleted.	High
FR-06	Record income or expense with amount, date, category and an optional note (max 200 characters).	High
FR-07	Edit or delete any transaction.	High
FR-08	List a month's transactions, newest first, grouped by day, filterable by type and searchable.	Medium
FR-09	Set, update and remove one monthly limit per category.	High
FR-10	Show spent, remaining and % used per budget, with status <i>On track</i> (<80%), <i>Close to limit</i> (80–100%) or <i>Over budget</i> (>100%).	High
FR-11	Show monthly income, expenses, balance and savings rate.	High
FR-12	Chart spending by category with each category's share.	High
FR-13	Chart income vs expenses for the last six months, including empty months.	Medium
FR-14	Export a month's transactions as CSV with amounts in rupees.	Medium
FR-15	Import up to 1,000 CSV rows; create missing categories; if any row is invalid, import nothing and list each bad line with its reason.	High

4. Non-Functional Requirements

ID	Quality	Requirement
NFR-01	Security	Passwords hashed with salted scrypt; random 256-bit session tokens stored only as hashes, sent in HttpOnly, SameSite cookies.
NFR-02	Security	Max 20 failed log-in/sign-up attempts per 15 minutes per client; same message for wrong email and wrong password.
NFR-03	Data integrity	Money stored as integer paisa; CSV import is all-or-nothing; foreign keys enforced.
NFR-04	Reliability	All input validated on the server; clear error messages; no crashes; no internal details shown.
NFR-05	Performance	With 10,000 transactions: each request under 500 ms, dashboard under 2 s.
NFR-06	Usability	A transaction can be recorded in under 30 seconds (shortcut N); every action gives visible feedback.
NFR-07	Accessibility	Works from 360 px to desktop; keyboard usable; WCAG AA contrast (4.5:1); respects reduced motion.
NFR-08	Maintainability	Layered server (routes, controllers, services, repositories); automated tests pass before every commit.

ID	Quality	Requirement
NFR-09	Portability	Installs with npm on Windows, macOS and Linux; no database server needed.

5. Use Case Diagram

One actor: the **User**. All use cases except *Sign up* and *Log in* need a session. «extend» runs only when spending reaches 80% of a budget.

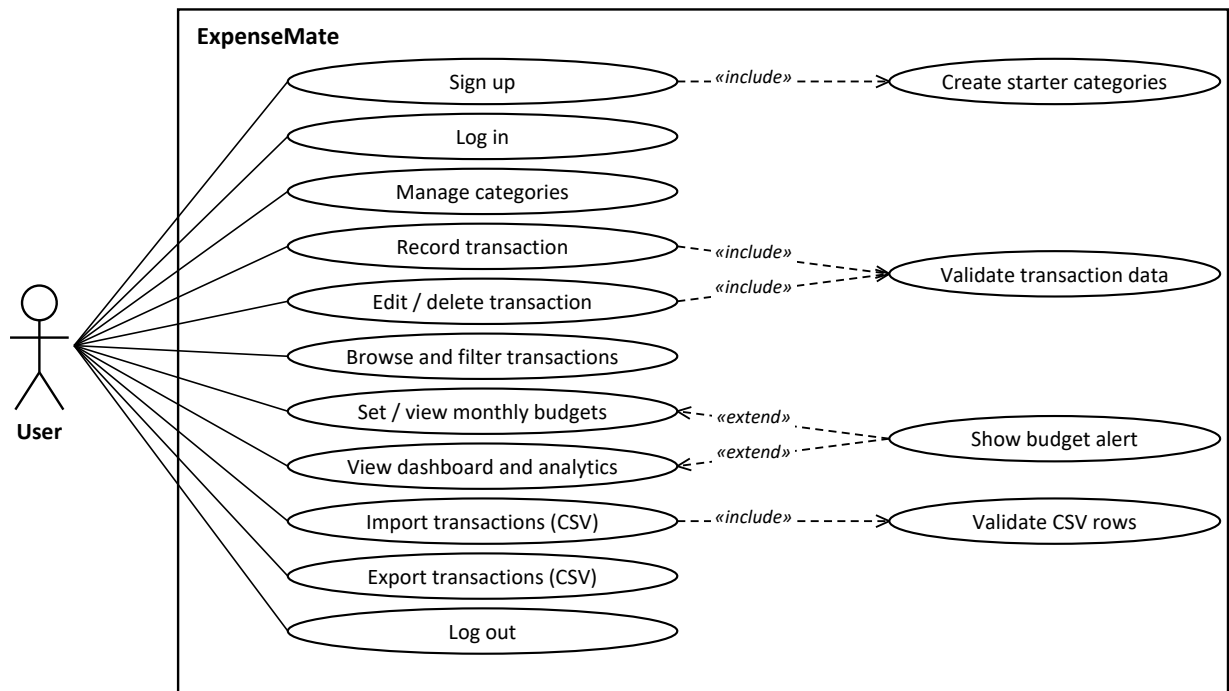


Figure 1. Use case diagram of ExpenseMate